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Math 400

Section 2.4 The Precise Definition of a Limit

2 Precise Definition of a Limit Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then we say that the **limit of $f(x)$ as x approaches a is L**, and we write

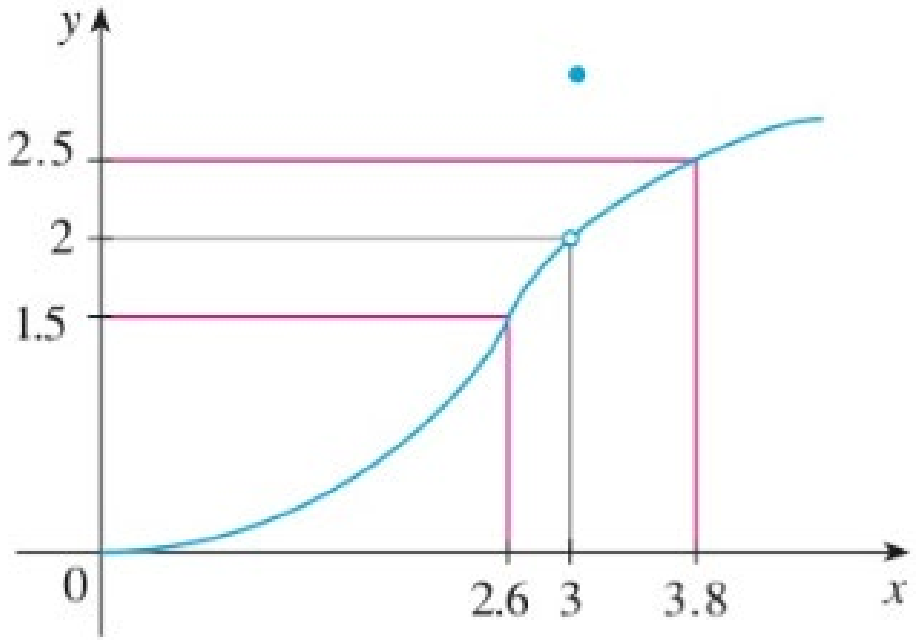
$$\lim_{x \to a} f(x) = L$$

if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$\text{if} \quad 0 < |x - a| < \delta \qquad \text{then} \qquad |f(x) - L| < \varepsilon$$

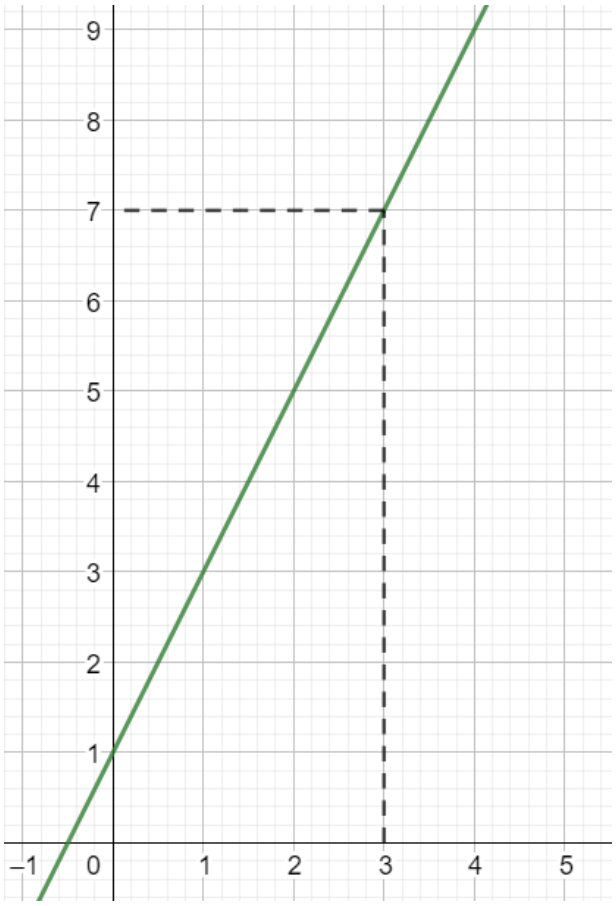
Example 1: Use the given graph of f to find a number δ such that

If $0 < |x - 3| < \delta$ then $|f(x) - 2| < 0.5$. (Section 2.4 page 113 #2)



Example 2: Find the following limit.

$$\lim_{x \to 3}(2x + 1)$$



Example 3: Prove the statement using the ε, δ definition of a limit.

$\lim_{x \to 3}(2x + 1) = 7$

Steps for proving that $\lim_{x \to a} f(x) = L$

1. **Find δ.** Let ε be an arbitrary positive number. Use the inequality $|f(x) - L| < \varepsilon$ to find a condition of the form $|x - a| < \delta$, where δ depends only on the value of ε.
2. **Write a proof.** For any $\varepsilon > 0$, assume $0 < |x - a| < \delta$ and use the relationship between ε and δ found in Step 1 to prove that $|f(x) - L| < \varepsilon$.

Example 3: (continued)

Example 4: Prove the statement using the ε, δ definition of a limit.

$$\lim_{x \to -2} (5x + 4) = -6$$

Example 4: (continued)

Example 5: Prove the statement using the ε, δ definition of a limit.

$\lim_{x \to 10} (-8x + 54) = -26$

Steps for proving that $\lim_{x \to a} f(x) = L$

1. **Find δ.** Let ε be an arbitrary positive number. Use the inequality $|f(x) - L| < \varepsilon$ to find a condition of the form $|x - a| < \delta$, where δ depends only on the value of ε.
2. **Write a proof.** For any $\varepsilon > 0$, assume $0 < |x - a| < \delta$ and use the relationship between ε and δ found in Step 1 to prove that $|f(x) - L| < \varepsilon$.

Example 5: (continued)

Example 6: Prove the statement using the ε, δ definition of a limit.

$$\lim_{x \to -2} \frac{5x^2 + 4x - 12}{x + 2} = -16$$

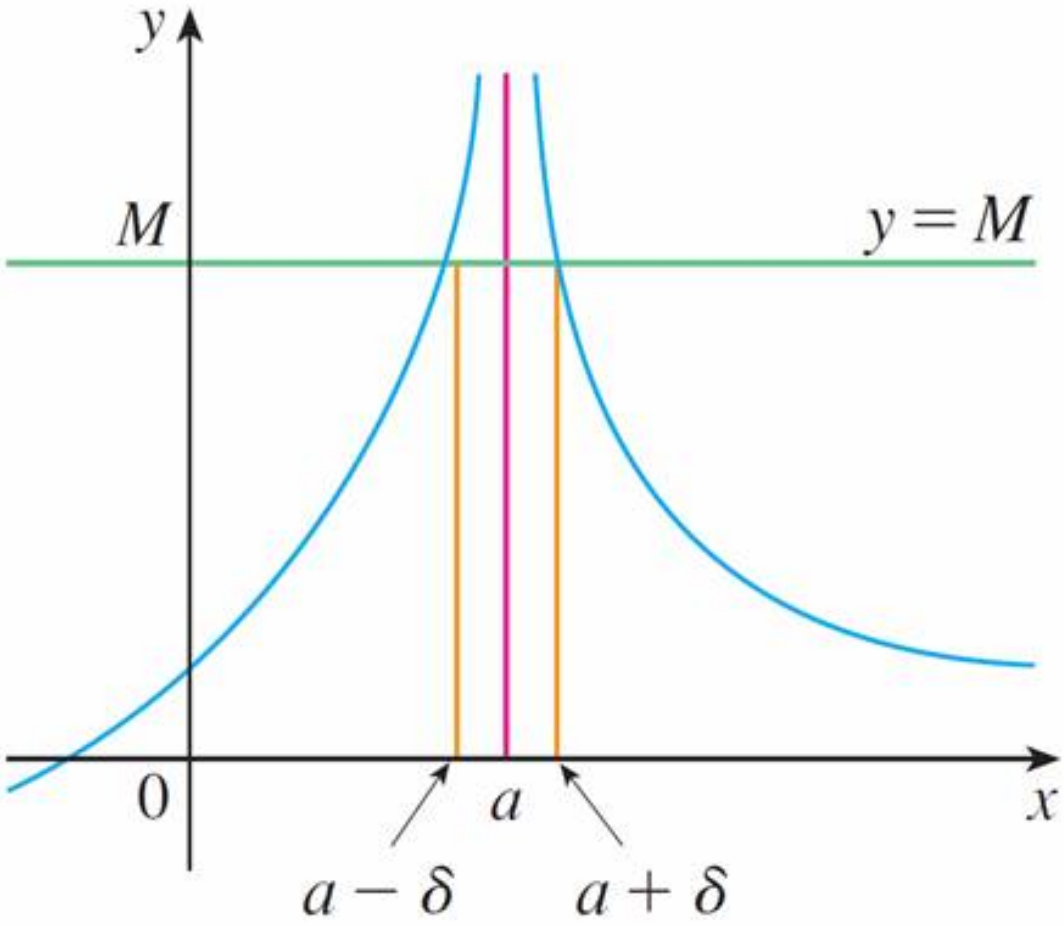
Example 6: (continued)

6 Precise Definition of an Infinite Limit Let f be a function defined on some open interval that contains the number a, except possibly at a itself. Then

$$\lim_{x \to a} f(x) = \infty$$

means that for every positive number M there is a positive number δ such that

$$\text{if} \quad 0 < |x - a| < \delta \quad \text{then} \quad f(x) > M$$


y
M
y = M
0
a
x
a − δ
a + δ

Example 7: Use the precise definition of a limit to prove the following limit.

$$\lim_{x \to 0} \frac{1}{x^2} = \infty$$

Steps for proving that $\lim_{x \to a} f(x) = \infty$

1. Find δ. Let M be an arbitrary positive number. Use the statement $f(x) > M$ to find an inequality of the form $|x - a| < \delta$, where δ depends only on M.

2. For any $M > 0$, assume $0 < |x - a| < \delta$ and use the relationship between M and δ found in Step 1 to prove that $f(x) > M$.

Example 7: (continued)

Example 8: Use the precise definition of a limit to prove the following limit.

$$\lim_{x\to 1} \frac{1}{(x-1)^2} = \infty$$

Steps for proving that $\lim_{x\to a} f(x) = \infty$

1. Find δ. Let M be an arbitrary positive number. Use the statement $f(x) > M$ to find an inequality of the form $|x - a| < \delta$, where δ depends only on M.

2. For any $M > 0$, assume $0 < |x - a| < \delta$ and use the relationship between M and δ found in Step 1 to prove that $f(x) > M$.

Example 8: (continued)